

(PROJECT PROPSAL)



Line-Following Robot with 2 Sensors (version 1)

Robot avoiding obstacles (version 2)

Vasawat prapachaimongkol

Peeradon buranathawon

Apichai udomsin

Thanapol kanpa

**BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING**

MAE FAH LUANG UNIVERSITY

2023

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A COMPUTER ENGINEERING PROJECT SUBMITTED TO
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING

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THIS COMPUTER ENGINEERING PROJECT HAS BEE APPROVED
TO BE A PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE DEGREE OF BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING

2022

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ACKNOWLEDGEMENT

we would like to thank our project supervisor, who provided us with invaluable guidance and support throughout the project. Their expertise and experience were instrumental in ensuring the project's success.

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Vasawat prapachaimongkol
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Abstract

The "Robot Avoiding Obstacles" project is a comprehensive exploration of robotics and automation, focusing on the development of a robotic system capable of autonomously detecting and circumventing obstacles in its path. The primary goal of this project is to design, construct, and program a versatile robot equipped with sensor technology to ensure safe navigation in dynamic environments.

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6. 1 Channel TCRT5000 Tracker Sensor
7. M2.5 x 12mm male-female brass standoff screws and nuts
8. 18650 battery holder (2-cell)
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6. 1 Channel TCRT5000 Tracker Sensor
7. M2.5 x 12mm male-female brass standoff screws and nuts
8. 18650 battery holder (2-cell)
9. HY-SRF05 SRF05 Ultrasonic Distance Sensor
10. Mounting Bracket for HC-SR04

Chapter1 INTRODUCTION

1.1 Background and Rational

Version 1

The use of robotics technology in various industries has increased significantly in recent years. One area where robotics technology has proven useful is in automating specific tasks, such as line-following. Line-following robots are designed to autonomously follow a line on the ground, making them useful in industries such as manufacturing and logistics. This project aims to develop a line-following robot with 2 sensors that can perform this task autonomously.

Version 2

Automation and robotics are becoming integral in various industries. Ensuring robots can navigate obstacles autonomously is crucial for safety and efficiency, driven by advancements in sensor technology, the "Robot Avoiding Obstacles" project addresses the need for safe, efficient, and technologically advanced autonomous robotic systems, offering practical solutions while promoting learning and innovation.

1.2 Objective

Version 1

The objective of this project is to design, develop and test a line-following robot with 2 sensors that can autonomously follow a line on the ground. The robot should be able to continuously detect the color of the surface beneath it and adjust its movements in real-time to stay on course. The project aims to demonstrate the application of robotics technology in automating specific tasks in industries such as manufacturing and logistics.

Version 2

the primary objective of the "Robot Avoiding Obstacles" project is to design, build, and program a robot that can autonomously navigate its environment while detecting and avoiding obstacles, thereby ensuring safe and efficient movement in dynamic surroundings.

1.3 Scope

Version 1

1.3.1 The robot should be capable of color detection

1.3.2 The robot should be capable of turning left, turning right, and moving forward.

1.3.3 The robot should be capable of moving along a predefined path

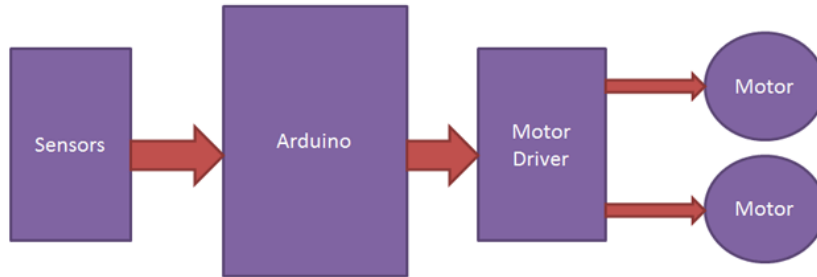
Version 2

1.3.1 The robot should be capable of color detection.

1.3.2 Zxczxczxc

1.3.3 zxczxczxc

1.4 System design



1.5 Methodology

1.5.1 Proposal preparation and defense

1.5.2 Literature review and requirement gathering

1.5.3 System analysis

1.5.4 System design

1.5.5 Senior project 1 document preparation

1.5.6 Senior Project 1 exam

1.5.7 System development

1.5.8 System testing

1.5.9 Senior Project 2 exam

1.5.10 Senior Project 2 exam

1.6 Plan

Table 0.1 Working plan

	Senior project 1					Senior project 2				
Plan/Month	Jan	Feb	Mar	Apr	May	Aug	Sep	Oct	Nov	Dec
Proposal preparation and defense										
Literature review and requirement gathering										
System analysis										
System design										
Senior project 1 document preparation										
Senior Project 1 exam										
System development										
System testing										
Senior project 2 document preparation										
Senior Project 2 exam										

1.7 Expected Result

Version 1

The expected result of this project is a line-following robot with 2 sensors that can autonomously follow a line on the ground. The robot should be able to continuously detect the color of the surface beneath it and adjust its movements in real-time to stay on course.

Version 2

These expected results outline the project's goals and outcomes, demonstrating the successful creation of a robot capable of avoiding obstacles while navigating autonomously in dynamic environments.

1.8 Place and Equipment (version 1)

1.8.1 Place (in my dormitory)

1.8.2 Equipment hardware

2WD Smart Car Robot Chassis Kits
Arduino UNO R3 - Made in Italy
L298P Motor Shield Board
Jumper (M2M) cable 20cm Male to Male
Jumper (F2F) cable 30cm Female to Female
18650 Battery Holder, 2 slots
3.7v Rechargeable Battery, 2 pieces
1 Channel TCRT5000 Tracker Sensor, 2 pieces
Metal PCB Board Support, 25mm long.

1.8.3 Equipment software

Arduino

1.9 Place and Equipment (version 2)

1.9.1 Place (in my dormitory)

1.9.2 Equipment hardware

2WD Smart Car Robot Chassis Kits
Arduino UNO R3 - Made in Italy
L298P Motor Shield Board
Jumper (M2M) cable 20cm Male to Male
Jumper (F2F) cable 30cm Female to Female
18650 Battery Holder, 2 slots
3.7v Rechargeable Battery, 2 pieces
HY-SRF05 SRF05 Ultrasonic Distance Sensor
Metal PCB Board Support, 25mm long.

1.9.3 Equipment software

Arduino

Chapter 2 LITERATURE REVIEW

2.1 Related Theory

The related theory for this project is the integration of several technologies, including sensors, motor control, and programming, to achieve the desired functionality of the Robot avoiding obstacles. The following are some key concepts and theories related to this project:

Sensor technology: a strong grasp of sensor technology, specifically the principles and characteristics of the HY-SRF05 SRF05 Ultrasonic Distance Sensor, is essential for the successful implementation of obstacle detection and avoidance in the "Robot Avoiding Obstacles" project. This sensor plays a pivotal role in providing data that enables the robot to detect and respond to obstacles in its path.

Motor control: Motor control is essential for the robot's movement. The robot will use a motor shield board to control the speed and direction of the motors. The microcontroller will send signals to the motor shield board to adjust the motor speed and direction based on the feedback from the sensors.

Programming: Programming is critical for the robot to perform the desired functions. The robot will be programmed using an Arduino IDE to read data from the sensors and control the motors. The programming will use a combination of conditional statements and loops to control the movement of the robot.

2.2 Related Research

Research different ways robots can dodge things in their path. It's like finding the best path through a crowded hallway

Ultrasonic Sensors: Learn how ultrasonic sensors (like the HY-SRF05) work for detecting objects. It's like understanding how bats use sound to find their way.

Smart Robots: Explore how robots can be smart like humans. Think about how they decide when to go left, right, or stop when they see an obstacle, just like you do when you avoid obstacles in your way.

Robot Safety: Look into how robots avoid bumping into people or things. It's like learning how to drive a car safely.

Open-Source Robots: Check out cool robot projects online. There are lots of people who share their robot-building secrets, just like sharing DIY videos on YouTube.

Robots in the Real World: See how robots help with tasks like cleaning floors (like a Roomba), delivering packages, or even flying drones. It's like learning how technology makes our lives easier.

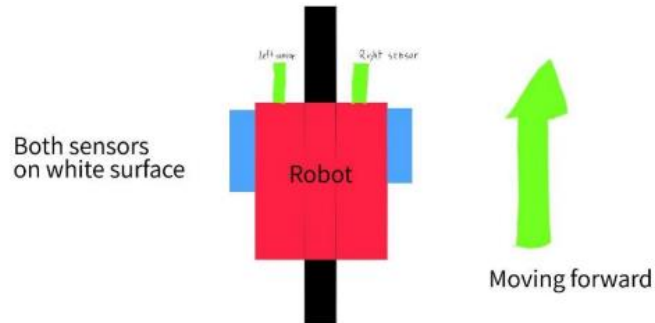
Robot Kits and Components: Explore how the robot kit components you have, like the Arduino and ultrasonic sensor, fit into the big picture of building your robot. Think of it as putting together a complex LEGO set.

Remember, research is like detective work. You're trying to find clues and ideas from others who've already explored the same topics to help you build an awesome obstacle-avoiding robot for your project!.

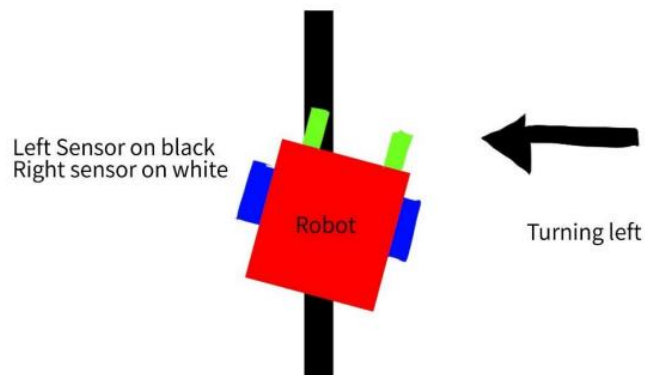
2.3 Related Work

Version 1

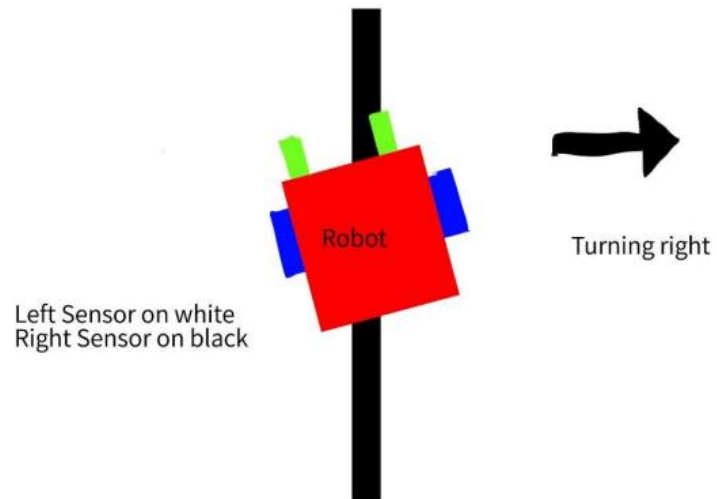
In this Arduino + L298P line following robot project, we will use 2 sensors, left side and right side. Both left and right Checked, it's a white ground. The beam can be reflected back both (1 , 1) for the robot to move forward.



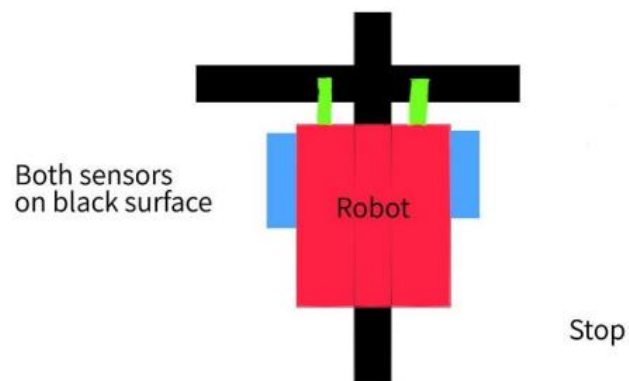
If the left sensor meets the black ground and the right sensor meets the white ground (0, 1), the robot turns to the left.



If the right sensor meets the black ground and the left sensor meets the white ground (1, 0), the robot turns to the right.



If both sensors come on both black lines (0, 0), the robot stops.



Version 2

The HY-SRF05 distance measuring sensor uses ultrasonic sound waves to work. The principle is that when the transmitter sends out a sound, the sound hits an object and the wave is reflected back, and the receiver receives it. The time measured after sending and receiving back is calculated by comparing it with the speed of sound to obtain the distance.

The operation starts from triggering the signal into the Trig pin to be HIGH, which will cause the module to start measuring distance. Then send the measured value as a pulse width to the Echo pin. Take the time value of the pulse width sent from the Echo pin and divide it by $29 / 2$ to get the distance value in centimeters. It can measure distances from 2 centimeters to 4.5 meters. Errors depend on the distance measured.

2.4 Related Technology

Version 1

Hardware

1 Channel TCRT5000 Tracker Sensor: It is a type of infrared sensor that is commonly used in line-following robots. It consists of an infrared LED and a phototransistor that are positioned facing each other. When the sensor is placed over a black line, the infrared light is absorbed, and the phototransistor detects a decrease in light intensity. This allows the robot to detect the line and follow it.



Version 2

Software



C++

C++ is a new widely used computer programming language. It was developed from the C language. It is a general-purpose programming language that supports multiple programming paradigms, such as Procedural programming, Abstraction, Object-oriented programming, generic programming, and design pattern. This language has faster performance compared to other languages and is easy to use. It is popularly used in Operating Systems, Games, GUI Business, Web Browsers, Embedded Systems, Banking Applications, Compilers, Database Managements Softwares, Cloud/Distributed Systems, Libraries, Switches, Advanced Computation Graphics. In our project, C++ will use in Arduino IDE



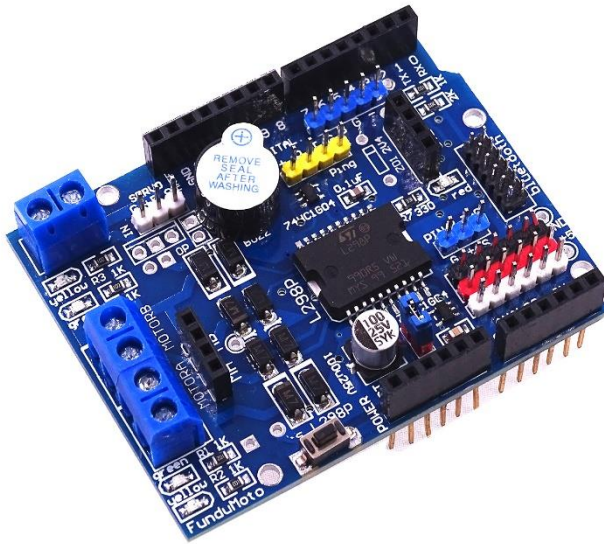
Arduino IDE

The Arduino IDE is an open-source software, which is used to write sketch code, verify/compile, and upload code to microcontroller on the Arduino boards. The Arduino can show the data in both numeric data and line graphs. At first Arduino IDE can be used with Arduino boards only, but now it can be used with other boards and Node MCU. The IDE application is suitable for different operating systems such as Windows, Mac OS X, and Linux. It supports the programming languages C and C++.

Hardware

L298P Motor Shield Board

: It is a motor driver board that is used to control the speed and direction of DC motors. It can control up to two DC motors simultaneously and is compatible with the Arduino UNO board. It provides a high voltage and current capacity and is commonly used in robotics projects.



HY-SRF05 SRF05 Ultrasonic Distance Sensor Module

: Ultrasonic distance measurement module It communicates with the microcontroller through triggering signals. Can measure distances from 2 centimeters to 4 meters. Uses a voltage of 5V to measure the distance between intervening objects. with the sensor face

Used to communicate with the microcontroller through triggering signals at the Trig pin.

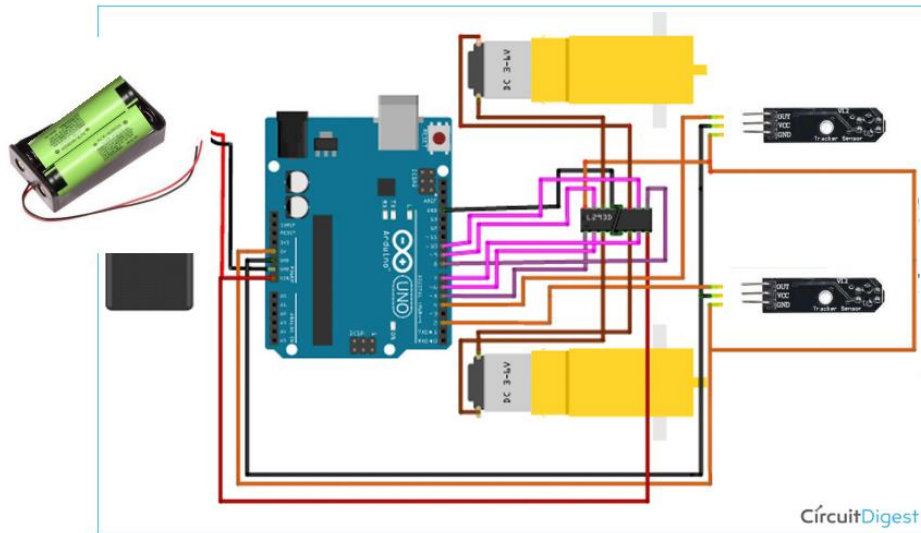
Can measure distances from 2 centimeters to 4.5 meters. Errors depend on the distance measured. Use a voltage of 5V and use a current of <2mA.



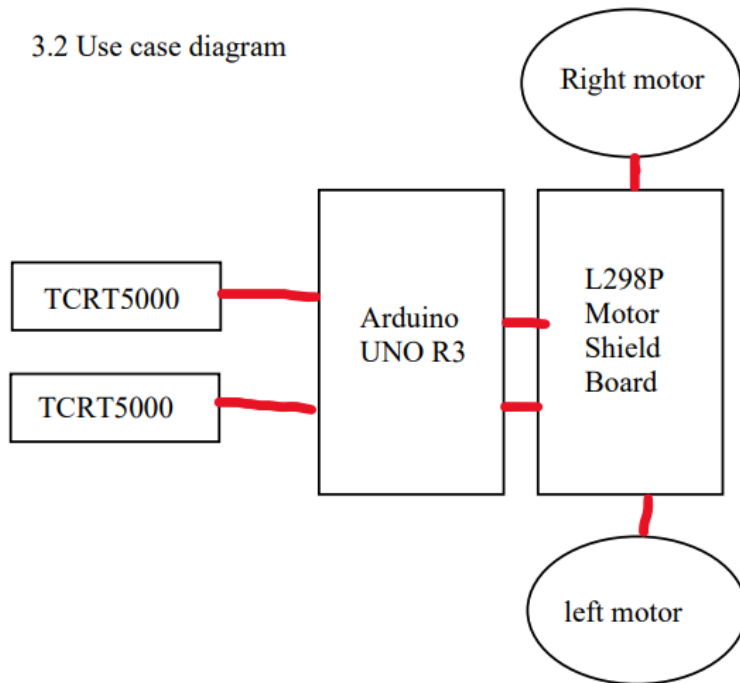
Chapter 3 ANALYSIS AND DESIGN

Version 1

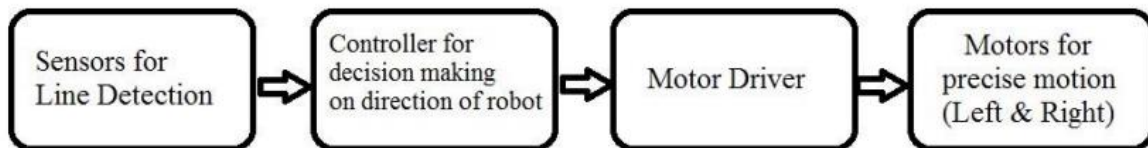
3.1 circuit design



3.2 Use case diagram



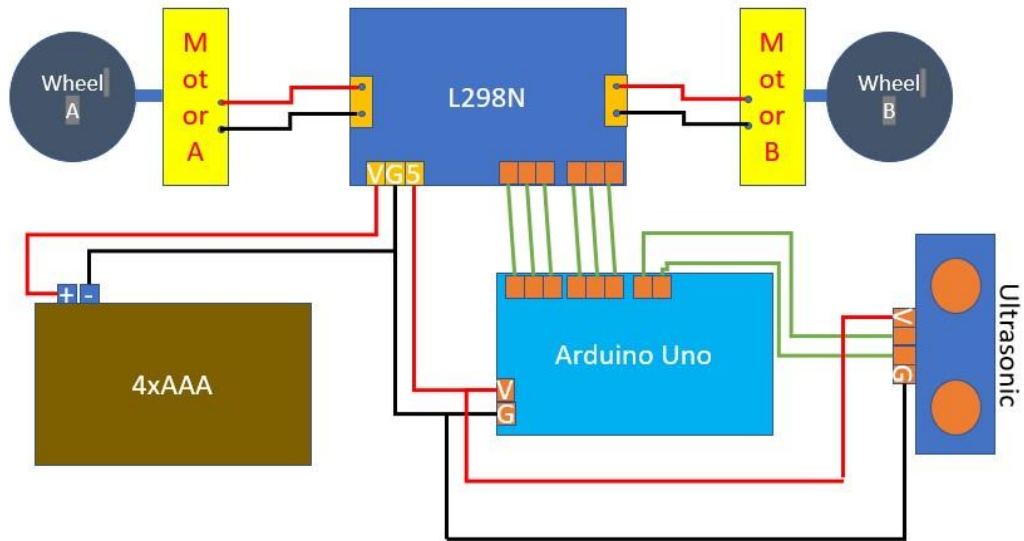
3.2 Activity diagram



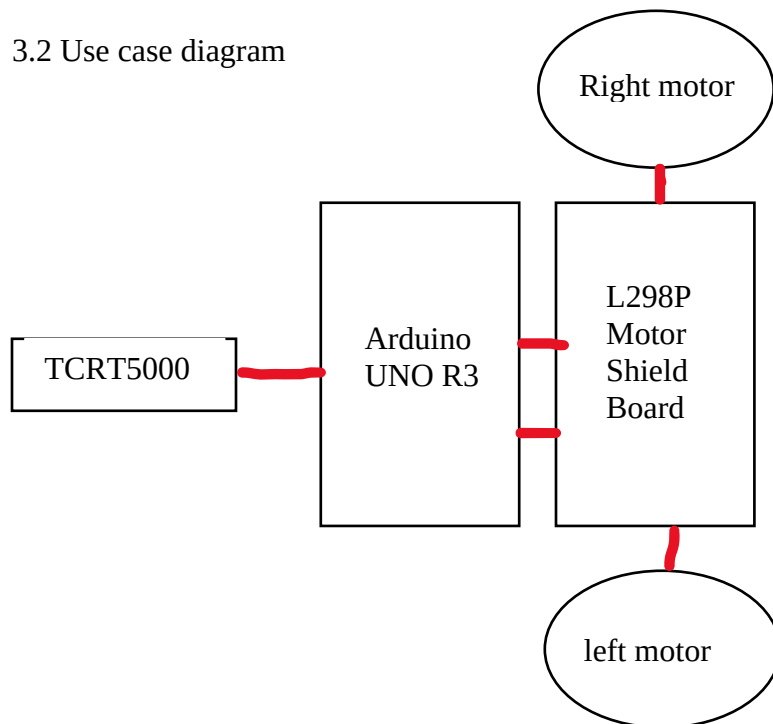
Block Diagram for Line Follower Robot

Version 2

3.1 circuit design



3.2 Use case diagram



Coding (version 2)

-Backward

```
void Backward(int time)
{
    digitalWrite(MA1, LOW);
    digitalWrite(MA2, HIGH);
    analogWrite(PWM_A, SPEED);

    digitalWrite(MB1, HIGH);
    digitalWrite(MB2, LOW);
    analogWrite(PWM_B, SPEED);

    delay(time);
}
```

-Forward

```
void Forward (int time)
{
    digitalWrite(MA1, HIGH);
    digitalWrite(MA2, LOW);
    analogWrite(PWM_A, SPEED);

    digitalWrite(MB1, LOW);
    digitalWrite(MB2, HIGH);
    analogWrite(PWM_B, SPEED);

    delay(time);
}
```

-Turnleft

```
void turnLeft(int time)
{
    digitalWrite(MA1, HIGH);
    digitalWrite(MA2, LOW);
    analogWrite(PWM_A, SPEED);

    digitalWrite(MB1, LOW);
    digitalWrite(MB2, LOW);
    analogWrite(PWM_B, 0);

    delay(time);
}
```

-Turnright

```
void turnRight(int time)
{
    digitalWrite(MA1, LOW);
    digitalWrite(MA2, LOW);
    analogWrite(PWM_A, 0);

    digitalWrite(MB1, LOW);
    digitalWrite(MB2, HIGH);
    analogWrite(PWM_B, SPEED);

    delay(time);
}
```

-Stop

```
void Stop(int time)
{
    digitalWrite(MA1, LOW);
    digitalWrite(MA2, LOW);
    analogWrite(PWM_A, 0);

    digitalWrite(MB1, LOW);
    digitalWrite(MB2, LOW);
    analogWrite(PWM_B, 0);

    delay(time);
}
```

-Main function used to check distance

```
void setup() {

    pinMode(4, OUTPUT); //Buzzer

    //Setup Channel A
    pinMode(12, OUTPUT); //Motor A1
    pinMode(3, OUTPUT); //Motor A2
    pinMode(10, OUTPUT); //Speed PWM Motor A

    //Setup Channel B
    pinMode(13, OUTPUT); //Motor B1
    pinMode(8, OUTPUT); //Motor B2
    pinMode(11, OUTPUT); //Speed PWM Motor B

    pinMode(TRIG_PIN, OUTPUT);
    pinMode(ECHO_PIN, INPUT);
    Serial.begin(BAUD_RATE);

}

void loop() {

    digitalWrite(TRIG_PIN, LOW);

    delayMicroseconds(2);

    digitalWrite(TRIG_PIN, HIGH);

    delayMicroseconds(10);

    digitalWrite(TRIG_PIN, LOW);
```

```

const unsigned long duration = pulseIn(ECHO_PIN, HIGH);

int distance = duration / 29 / 2;

if ((distance > 1) && (distance < 20)) {

    digitalWrite(4, HIGH);
    Stop(500);
    digitalWrite(4, LOW);
    Backward(400);
    turnLeft(400);
    Serial.print("distance to nearest object:");
    Serial.println(distance);
    Serial.println(" cm");

}

else {

    digitalWrite(4, LOW);
    Forward(10);
    Serial.print("distance to nearest object:");
    Serial.println(distance);
    Serial.println(" cm");

}

delay(100);

}

```

Chapter 4 CONCLUSION

4.1 Conclusion: version 1

In conclusion, we have successfully designed and implemented a line-following robot with two sensors using the Arduino UNO R3, L298P motor shield board, TCRT5000 tracker sensors, and 18650 Panasonic NCR18650B 3.7v batteries. The robot is capable of detecting a black line on a white surface and following it while making turns and changing direction as required.

4.1.1 Discussion: version 1

Our experimental results show that our line-following robot with two sensors is able to follow a black line on a white surface with reasonable accuracy and efficiency. However, we did encounter some difficulties in calibration and programming, which required several iterations to fine-tune the robot's performance.

4.2 Conclusion: version 2

the robot avoiding obstacles project has laid a solid foundation for autonomous navigation systems. The successful implementation of obstacle detection and avoidance mechanisms, coupled with efficient path planning algorithms, showcases the potential for further advancements in the field. This project contributes to the ongoing efforts in creating intelligent, adaptable robotic systems capable of navigating real-world environments with precision and safety.

4.2.1 Discussion: version 2

Our experimental results show that our obstacle avoidance robot with ultrasonic sensors can dodge obstacles accurately and efficiently. However, we encountered some problems in calibration and programming. This needs to be repeated many times to fine-tune the robot's performance.

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<https://www.analogread.com/article/68/%E0%B8%AA%E0%B8%AD%E0%B8%99%E0%B9%83%E0%B8%8A%E0%B9%89%E0%B8%87%E0%B8%B2%E0%B8%99%E0%B8%9A%E0%B8%AD%E0%B8%A3%E0%B9%8C%E0%B8%94-arduino-%E0%B8%81%E0%B8%B1%E0%B8%9A-ultrasonic-module-hy-srf05-%E0%B8%9E%E0%B8%A3%E0%B9%89%E0%B8%AD%E0%B8%A1%E0%B9%82%E0%B8%84%E0%B9%89%E0%B8%94%E0%B8%95%E0%B8%B1%E0%B8%A7%E0%B8%AD%E0%B8%A2%E0%B9%88%E0%B8%B2%E0%B8%87%E0%B8%9B%E0%B8%A3%E0%B8%B0%E0%B8%A2%E0%B8%B8%E0%B8%81%E0%B8%95%E0%B9%8C%E0%B9%83%E0%B8%8A%E0%B9%89%E0%B8%87%E0%B8%B2%E0%B8%99>
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